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Enhancing Forensic Evidence Integrity Using Permissioned Blockchain Technology

Using Smart Contracts and Cryptographic Techniques for Secure Chain of Custody

[1]V. Akhila, [2]Bhavani Shankar, [3]Farhan,

[4] Guide: Bagam Laxmaiah

[1]Student, [2]Student, [3]Student, [4]Associate Professor

Department of Computer Science and Engineering,

CMR Technical Campus, Hyderabad, India

Abstract : Proper forensic evidence management is critical to ensure justice and the integrity of judicial processes. Forensic evidence collected from crime scenes must be protected against modification, unauthorized access, or loss throughout its entire life cycle. Inadequate management of the evidence chain of custody may cause evidence to be ruled inadmissible in court, which may lead to the dismissal of a case. Conventional evidence management systems are prone to tampering, data breaches, and human mistakes. To overcome these challenges, this paper proposes a blockchain-based framework for the digital management of forensic evidence while maintaining the integrity of the evidence chain of custody. The proposed system utilizes a permissioned blockchain model inspired by Hyperledger Fabric and incorporates Ethereum's ERC-721 token standard to assign a unique non-fungible token (NFT) to each piece of evidence. Hash functions are used to ensure data integrity, and smart contracts automate logging, verification, and auditing processes. Role-based access control and wallet authentication limit the use of the system to authorized personnel. A hybrid storage model is used, where critical metadata such as timestamps and hash values are stored on the blockchain, and large forensic files are stored securely off-chain. The proposed system improves the transparency, traceability, and security

IndexTerms - Blockchain, Forensic Evidence, Chain of Custody, Smart Contracts, Cryptographic Hashing

INTRODUCTION

The concept of managing the evidence is of utmost importance in the domain of forensics and the justice process. The integrity of the evidence through its chain of custody needs to be maintained right from its procurement till it is presented in the courtroom. The term 'Chain of Custody' represents the chronological record of handling an evidence without any modifications.

Existing systems, which include manual records or centralized digital records of evidence management, suffer from many drawbacks like data manipulation or errors and unapproved access to the information stored in the system. These problems can result in the invalidation of an investigation or rejection of evidence by the court.

This paper proposes the use of blockchain to improve the existing methods of managing forensic evidence.

NEED OF THE STUDY.

Forensic evidence management within criminal investigations must be properly done to preserve its integrity. Manual record keeping and database systems are likely to experience data manipulation, alteration, and corruption, thereby making the evidence less credible during trial processes.

Considering the increased cases of cybercrime, there is need for an improved forensic data management system that is reliable and secure. Blockchain technology is characterized by properties such as immutability, transparency, and decentralization; hence, it ensures a tamper-proof chain of custody.

This research therefore aims at designing a forensic evidence management system based on blockchain technology.

3.1 Population and Sample
A synthetic dataset will be used for forensic evidence data records. A total of 100–500 records are synthesized for various categories of crimes and evidence.

3.2 Data and Sources of Data
Python programming language will be used to synthesize the data. Hash values will be derived from SHA-256, and blockchain transactions will be used for testing purposes.

3.3 Theoretical Framework
Blockchain technology, smart contracts, and hashing algorithms will be used within the system framework.

RESEARCH METHODOLOGY

The methodology section describes the process and approach used in carrying out the research study. It involves datasets, sources of data, system components, and theoretical framework used. Details include:

3.1 Population and Sample
In this study, forensic evidence records will be considered the dataset. However, because the information in real-life forensic cases may be confidential, an artificially created dataset will be developed. There will be around 100-500 records for various crime types and evidence.

The sample will be made up of records with attributes such as Evidence ID, Case ID, Crime Type, Evidence Type, Collection Timestamp, Location, and Hash Value. These records will be evaluated in relation to the functionality of the proposed blockchain system.

3.2 Data and Sources of Data
Synthetic data will be used in this study. It will be developed using the Python programming language. Time-based data about forensic evidence collected and kept will be included.

There will be generated hash values from the SHA-256 algorithm and blockchain transaction ID and wallet address information.

3.3 Theoretical Framework
This study involves blockchain technology to ensure forensic evidence is securely managed. It makes use of cryptography, hashing algorithms, and smart contract techniques.

The independent components include blockchain system, smart

3.4 System Architecture and Implementation
The suggested system architecture uses a web interface, a Django application server, and an Ethereum blockchain. The smart contract implementation is done using Solidity language.

Two main functionalities performed by the system are:

setEvidenceDetails() → Evidence details are saved
getData() → Stored evidence is retrieved

A user is required to be authenticated before using the system. Cryptography with public and private keys is applied for secure access.

3.5 Working Procedure
The working methodology comprises these stages:

Collection of data by an authorized user
Conversion of the data into hashes by using SHA-256 algorithm
Execution of the smart contract to save the data
Saving the data in the blockchain with time stamps and transaction IDs
Retrieving the saved data

The above steps make sure that the data is kept secure and immutable.

3.6 Analysis
Security, transparency, and integrity of the data are taken into consideration when analyzing the system. Blockchain makes sure that the data once stored is never changed. With the help of smart contracts, verification becomes easy, eliminating human errors.

Hybrid data storage technique makes the system efficient and cost-effective.

IV. RESULTS AND DISCUSSION

4.1 Results of Descriptive Statics of Study Variables

Table 4.1: Descriptive Statics

Variable	Minimum	Maximum	Mean	Std. Deviation	Jarque-Bera Test	Sig
Transaction Time (sec)	1.2	3.8	2.5	0.8	1.45	0.48
Hash Generation Time (sec)	0.01	0.05	0.02	0.01	1.12	0.57
Data Retrieval Time (sec)	0.5	2.0	1.2	0.4	1.67	0.43
Storage Efficiency (%)	85	98	92	4.5	1.38	0.50

Table 4.1 illustrates the descriptive statistics such as mean, standard deviation, minimum, maximum and Jarque-Bera test with their significance values of system performance variable. It can be seen from the above-mentioned statistics that the average values of variables including transaction time, hash generation time, data retrieval time and storage efficiency are 2.5, 0.02, 1.2 and 92 respectively.

It can be concluded from Table 4.1 that the maximum value of these variables in system performance is 3.8 seconds for transaction time, 0.05 seconds for hash generation time, 2.0 seconds for data retrieval time and 98% for storage efficiency.

It can be seen that the standard deviation values of the variables prove that the data is moderately dispersed around mean values. Furthermore, the sixth column in Table 4.1 represents the Jarque-Bera test which is used to check the normal distribution of the data.

Hypotheses of the normal distribution:
H0: Data is normally distributed
H1: Data is not normally distributed

Therefore, it can be stated from Table 4.1 that all system variables are normally distributed at a level of 5%.

The descriptive statistics of Table 4.1 indicate that the system performance is stable and does not depend on the changeability of system variables.

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